HW #4

1. (a) $x(n) = (-3)^n u(-n-2)$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n} = \sum_{n=-\infty}^{-2} (-3)^n z^{-n} \xrightarrow{m=-n} \sum_{m=2}^{\infty} (-3)^{-m} z^m = \sum_{m=2}^{\infty} \left(-\frac{z}{3}\right)^m = \frac{\left(-\frac{z}{3}\right)^2}{1-\left(-\frac{z}{3}\right)} = \boxed{\frac{z^2}{9+3z}}$$

with ROC: $\left|\frac{z}{3}\right| < 1 \Rightarrow |z| < 3$

(b) $x(n) = \begin{cases} 1 & -5 \le n \le 5 \\ 0 & \text{otherwise} \end{cases}$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n} = \sum_{n=-5}^{5} z^{-n} = \sum_{n=-5}^{5} \left(\frac{1}{z}\right)^n = \frac{\left(\frac{1}{z}\right)^{-5} - \left(\frac{1}{z}\right)^{5+1}}{1-\frac{1}{z}} = \frac{z^5 - z^{-6}}{1-\frac{1}{z}} = \boxed{\frac{z^{11}-1}{z^6 - z^5} \text{ with } z \neq 0 \text{ and } z \neq 1}$$

(c) $x(n) = \begin{cases} 4^{-n} & n \ge 1 \\ 2^n & n < 1 \end{cases}$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n} = \sum_{n=-\infty}^{0} 2^n z^{-n} + \sum_{n=1}^{\infty} 4^{-n} z^{-n} = \sum_{m=0}^{\infty} \left(\frac{z}{2}\right)^m + \sum_{n=1}^{\infty} \left(\frac{1}{4z}\right)^n = \frac{1}{1-\frac{z}{2}} + \frac{\frac{1}{4z}}{1-\frac{1}{4z}}$$

$$= \boxed{\frac{2}{2-z} + \frac{1}{4z-1}}$$

with ROC: $\left|\frac{z}{2}\right| < 1 \Rightarrow |z| < 2$ and $\left|\frac{1}{4z}\right| < 1 \Rightarrow |z| > \frac{1}{4}$

$\Rightarrow \frac{1}{4} < |z| < 2$

(d) $x(n) = [2, 1, 3, 5, -1, 6]$ (↑ at 3)

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n} = \boxed{2z^2 + z + 3 + 5z^{-1} - z^{-2} + 6z^{-3}}$$

2. (a) $x(n) = [n^2 2^{-n} + (n-1)3^{-n}]u(n)$

From $x(n) \longleftrightarrow X(z)$

$n x(n) \longleftrightarrow -z \cdot \frac{dX(z)}{dz}$

$n^2 x(n) \longleftrightarrow -z^2 \frac{dX(z)}{dz}$

$a^n u(n) \longleftrightarrow \frac{z}{z-a}$

$2^{-n} u(n) \longleftrightarrow \frac{z}{z-\frac{1}{2}} \Rightarrow \frac{d}{dz}\left(\frac{z}{z-\frac{1}{2}}\right) = \frac{\left(z-\frac{1}{2}\right) - z}{\left(z-\frac{1}{2}\right)^2} = \frac{-\frac{1}{2}}{\left(z-\frac{1}{2}\right)^2}$

$$\frac{d^2}{dz}\left(\frac{z}{z-\frac{1}{2}}\right) = \frac{d}{dz}\left[\frac{-\frac{1}{2}}{\left(z-\frac{1}{2}\right)^2}\right] = \frac{0 + \frac{1}{2} \cdot 2\left(z-\frac{1}{2}\right)}{\left(z-\frac{1}{2}\right)^4} = \frac{1}{\left(z-\frac{1}{2}\right)^3}$$

$n^2 2^{-n} u(n) \longleftrightarrow -z^2 \cdot \frac{1}{\left(z-\frac{1}{2}\right)^3} = \frac{-z^2}{\left(z-\frac{1}{2}\right)^3}$

$3^{-n} u(n) \longleftrightarrow \frac{z}{z-\frac{1}{3}}$

$n 3^{-n} u(n) \longleftrightarrow -z \cdot \frac{d}{dz}\left(\frac{z}{z-\frac{1}{3}}\right) = -z \cdot \frac{z - \left(z-\frac{1}{3}\right)}{\left(z-\frac{1}{3}\right)^2} = \frac{\frac{1}{3}z}{\left(z-\frac{1}{3}\right)^2}$

So: $\boxed{X(z) = \frac{-z^2}{\left(z-\frac{1}{2}\right)^3} + \frac{\frac{1}{3}z}{\left(z-\frac{1}{3}\right)^2} - \frac{z}{z-\frac{1}{3}} \text{ with ROC: } |z| > \frac{1}{2}}$

(b). $x(n) = 2e^{-n}\cos(\omega n)u(n)$

From $\cos(\omega n) = \frac{1}{2}\left[e^{j\omega n} + e^{-j\omega n}\right]$, so that $x(n) = e^{-n}\left[e^{j\omega n} + e^{-j\omega n}\right]u(n)$

$$= e^{(j\omega-1)n}u(n) + e^{-(j\omega+1)n}u(n)$$

$$\Rightarrow X(z) = \frac{z}{z - e^{j\omega-1}} + \frac{z}{z - e^{-(j\omega+1)}}$$

$$= \frac{z\cdot\left[z - e^{-(j\omega+1)}\right] + z\cdot\left(z - e^{j\omega-1}\right)}{(z - e^{j\omega-1})(z - e^{-(j\omega+1)})}$$

$$= \frac{z^2 - z\cdot e^{-(j\omega+1)} + z^2 - z\cdot e^{j\omega-1}}{z^2 - z\cdot e^{-(j\omega+1)} - z\cdot e^{j\omega-1} + e^{j\omega-1}\cdot e^{-(j\omega+1)}}$$

$$= \frac{2z^2 - ze^{-1}(e^{-j\omega} + e^{j\omega})}{z^2 - ze^{-1}(e^{-j\omega} + e^{j\omega}) + e^{-2}}$$

$$= \boxed{\frac{2z^2 - 2ze^{-1}\cos\omega}{z^2 - 2ze^{-1}\cos\omega + e^{-2}}}$$

with ROC: $\boxed{|z| > e^{-1}}$

3. $y(n) - y(n-1) + y(n-2) = 2^{-n}u(n)$

From: $y(n) \longleftrightarrow Y(z)$

$$y(n-1) \longleftrightarrow z^{-1}Y(z) + z^{-1}\sum_{m=-1}^{-1} y(m)z^{-m} = z^{-1}Y(z) + z^{-1}y(-1)\cdot z^{1} = z^{-1}Y(z) + 1$$

$$y(n-2) \longleftrightarrow z^{-2}Y(z) + z^{-2}\sum_{m=-2}^{-1} y(m)z^{-m} = z^{-2}Y(z) + z^{-2}\left[y(-2)z^2 + y(-1)z\right]$$

$$= z^{-2}Y(z) + 2 + z^{-1}$$

$$2^{-n}u(n) \longleftrightarrow \frac{z}{z-\frac{1}{2}}$$

$$Y(z) - \left[z^{-1}Y(z) + 1\right] + z^{-2}Y(z) + z^{-1} + 2 = \frac{z}{z-\frac{1}{2}}$$

$$Y(z) - z^{-1}Y(z) + z^{-2}Y(z) + z^{-1} + 1 = \frac{z}{z-\frac{1}{2}}$$

$$Y(z)\left[1 - z^{-1} + z^{-2}\right] = \frac{z}{z-\frac{1}{2}} - 1 - z^{-1}$$

$$Y(z) = \frac{\frac{z}{z-\frac{1}{2}} - 1 - z^{-1}}{1 - z^{-1} + z^{-2}} = \frac{\frac{z^3}{z-\frac{1}{2}} - z^2 - z}{z^2 - z + 1}$$

$$= \frac{z^3 - z^2\left(z-\frac{1}{2}\right) - z\left(z-\frac{1}{2}\right)}{(z^2 - z + 1)\left(z - \frac{1}{2}\right)} = \frac{\frac{1}{2}\left(z - z^2\right)}{\left(z^2 - z + 1\right)\left(z - \frac{1}{2}\right)}$$

$$\Rightarrow \boxed{Y(z) = \frac{z - z^2}{(z^2 - z + 1)(2z - 1)}}$$